5-DAY DRILL PACK · NETFLIX · STUDIO/CONTENT DE

The five days before the *conversation*.

Tuned for Sr/Staff Studio/Content. Five days, two-and-a-half hours per day, four weak areas. The point isn't volume — it's getting the same patterns through your fingers enough times that your hands type them while your mouth narrates trade-offs.

CONTENTS

§ 01 — THE DOMAIN

Studio/Content data is *not* Member/Growth data.

If you walk in expecting to talk about viewing events and DAU, you'll mismatch the interviewer's mental model. Studio/Content sits between the production process, the content catalog, the talent system, and finance. The data is SCD-heavy, multi-currency, hierarchical, and slow-moving in row count but high-stakes in correctness.

PRODUCTION LIFECYCLE

Greenlight → development → pre-production → production → post → QC → release → performance. Each stage has its own facts (deals signed, budget allocated, episodes shot, edit milestones, dub/sub completion). Time-to-stage and stage-cycle-time are the analytics.

TALENT & DEALS

Actors, directors, writers, crew. Hierarchical contracts (overall deal → series deal → episode fee). Royalty waterfalls. Residuals. Multi-party splits. SCD2 everywhere because deal terms are renegotiated.

CONTENT CATALOG

Title hierarchy (franchise → series → season → episode). Multi-region availability windows. Per-country ratings and certifications. Localization assets (audio, subtitle, key art) per language. Genre/mood/theme taxonomies that change.

PERFORMANCE ATTRIBUTION

Studio cares about: did our investment pay off? "Member value per dollar spent" is the holy-grail metric. This requires joining production cost data to viewing performance — and viewing has to be normalized for catalog effect (a title in 100M households has a head start over one in 10M).

FINANCE & ROYALTIES

Royalty calc by territory by license window. Currency conversion (multi-currency facts, FX rates as a dim). Amortization schedules (a show's cost is amortized over its expected lifetime — viewing pattern affects the amortization curve). This will come up.

THE UNIQUE VOCABULARY

"Slate," "greenlight," "pickup," "license window," "first-run," "library title," "tentpole," "originals vs licensed," "P&A" (prints & advertising), "residuals," "MFN" (most-favored-nation clause). Use these terms when modeling. Demonstrates context.

THE SR/STAFF DIFFERENTIATOR

At Sr/Staff level for Studio/Content, you're expected to think about **data products that influence greenlight decisions and renewal decisions** — not just dashboards. When asked "design a model for X," extend the answer to "...and here's what insight this unlocks for the studio leadership team." That framing alone lifts you a level.

§ 02 — 5-DAY PLAN

The plan, by the day.

~2.5 hours per day. Not negotiable: do the timed drills. The point isn't to find the answer — it's to get used to the clock running.

1 FOUNDATION · MODELING ~2.5 hrs

Modeling muscle memory.

30 MIN · READ

The five-move sequence

Re-read §02 of the main notebook. Practice saying the five moves out loud. Write each one on a sticky note next to your monitor. **Goal:** by end of week, you can recite them in 15 seconds.

60 MIN · DRILL

Modeling drills #1 and #2 (timed, 25 min each)

From §03 below: *Production lifecycle* and *Talent & royalties*. Use a real timer. Whiteboard, paper, or Excalidraw. Don't peek at the solution until the timer rings. After: compare yours to the worked answer, note what you missed.

30 MIN · VOCABULARY

Review the Studio/Content vocabulary above. Look up any terms you can't define crisply (P&A, MFN, residuals, slate, tentpole). Write 1-line definitions in your own words.

30 MIN · REFLECT

Did you ask clarifying questions before drawing tables? Did you state

grain explicitly? Did you volunteer two failure modes? If not — these are your reps for tomorrow.

2 SQL · WINDOW FUNCTIONS & GAPS/ISLANDS ~2.5 hrs

SQL fluency through repetition.

45 MIN · DRILL

SQL drills #1, #2, #3 (timed, 15 min each)

Solve at your own SQL playground or sqlfiddle. Time yourself. Don't just read the solutions — type them.

30 MIN · PATTERN REPS

Type out the gaps-and-islands template (LAG → flag → cumulative SUM) **three times from memory**. By rep three, your fingers should know it.

45 MIN · DRILL

SQL drills #4 and #5 (timed, 20 min each)

These are the harder ones — multi-CTE, finance-flavored. Talk out loud while solving (practice for the live round).

30 MIN · READ & REVIEW

Read §03 patterns 1, 2, 6 of the main notebook. Note the `QUALIFY` shortcut (Snowflake) — Netflix uses Snowflake heavily. Mention it in the round.

3 PYTHON · CODESIGNAL PACE ~2.5 hrs

Coding under a clock.

30 MIN · WARM UP

Type-from-memory templates

Open a blank file. Type out, from memory: (a) sessionization in Python, (b) top-K with heap, (c) sliding-window aggregator. These are the three patterns that cover most CodeSignal problems.

75 MIN · DRILL

Python drills #1, #2, #3 (timed, 25 min each)

Use a fresh editor (not your IDE – closer to CodeSignal feel). 25 min hard cap each. **Narrate while coding.** Record yourself if helpful – listen back to spot mumbling/long silences.

30 MIN · SOLVE AGAIN, OPTIMIZE

Take the slowest of your three solutions and rewrite it for better complexity. Practice the verbal arc: "My first pass was $O(n^2)$; I see I can use a hash map to get $O(n)$..."

15 MIN · REFLECT

Where did the clock catch you? Off-by-ones? Edge cases? Forgetting what `defaultdict(int)` does? Note one thing to drill tomorrow morning.

4 STREAMING · DEEP-DIVE PREP ~2.5 hrs

Vocabulary becomes fluency.

45 MIN · DRILL

Streaming concept Q&A (§06 below)

Read each prompt. **Answer out loud, on a timer (90 seconds each).** Then read the model answer. Goal: by end of week, the words "watermark," "exactly-once," "backpressure," "Kappa" come out without effort.

45 MIN · MOCK

Trace a play event from device to dashboard

Whiteboard or doc. 8 stages, 30 seconds each. Out loud. Repeat 3 times until it flows. This is the "narrate the architecture" question that almost always comes up in deep-dives.

30 MIN · RESUME REHEARSAL

Pick three projects from your resume. Write a 90-second STAR answer for each. Rehearse out loud. **Reminder per memory edits:** don't volunteer your specific employer, title, or founder role unless asked directly – describe the work.

30 MIN · CATCH-UP

Re-do any drill from days 1-3 you didn't finish. Or re-do the one you got wrong.

5 INTEGRATION · CALM

~2 hrs

Light reps, mental rehearsal, sleep.

30 MIN · ONE-PAGE REVIEW

Read §07 (cheat sheet) twice. Don't drill new material. Trust the reps you've put in.

30 MIN · ONE MOCK END-TO-END

Pick one modeling prompt + one SQL + one Python. 15 min each. Light pace, not a stress test. Confirm the muscle memory is there.

30 MIN · LOGISTICS

Test camera, mic, screen-share. Clear desktop. Have water ready. Have the recruiter's number open. Confirm timezone of the interview.

30 MIN · THE 3 QUESTIONS

Pick your three questions to ask each interviewer (from §08 of the main notebook). Write them on paper next to your monitor. Different questions for engineer vs manager.

THEN · STOP

No drills the night before. Walk, eat, sleep. The reps are already in. The interview is a conversation.

Five modeling prompts. *Whiteboard. Timer.*

Each prompt is sized for 25 minutes. Don't peek. Use the five-move sequence: clarify → grain → conceptual → physical → stress test. Whiteboard or paper is fine; type only if you're going to type during the real round.

MODEL · 01 PRODUCTION LIFECYCLE · 25 MIN

"Design a data model that lets the studio leadership team see, for any title in production, where it is in the pipeline (greenlight → development → pre-prod → production → post → QC → release) and how long it's been at each stage. They want to spot stuck projects."

▸ WALK-THROUGH

MODEL · 02 TALENT & ROYALTIES · 25 MIN

"Design a model for talent contracts and royalty payouts. An actor can have an overall deal with Netflix that covers multiple shows; within each show they have per-episode fees plus a backend (residuals, success bonuses). The legal team wants to be able to compute total liability per actor at any point in time, and finance wants to forecast next-quarter payouts."

▸ WALK-THROUGH

MODEL · 03 RENEWAL DECISIONS · 25 MIN

"Design a model that supports the renew/cancel decision for a series after season N has been released. The studio team wants a single 'renewal scorecard' per series with: total viewing, viewing per dollar spent, cost amortization status, talent renewal cost, and comparable shows."

▸ WALK-THROUGH

MODEL · 04 LOCALIZATION · 20 MIN

"Design a model for localization assets — every title needs dub and subtitle assets per language, each with its own QC state (in progress, in review, approved, rejected). The localization team needs to see which titles are blocking which territory launches."

▸ WALK-THROUGH

MODEL · 05 SLATE PLANNING · 25 MIN

"Design a model that supports the studio's annual slate planning — they need to see, for any future quarter, the expected mix of releases (genre, budget tier, target territory, talent tier) and how that compares to performance benchmarks of past slates. Help them spot gaps (e.g., 'we're under-indexed on family content in Q3')."

▸ WALK-THROUGH

§ 04 — SQL DRILLS

Five SQL drills. Type. Don't *read*.

These problems target your weak areas: window functions, gaps/islands, sessionization. All flavored for Studio/Content. Type each solution. Time-boxed. Compare to the model answer only after.

> THE SCHEMA YOU'RE WORKING WITH
>
> Assume tables: `dim_title(title_id, series_id, season_id, ep_num, runtime_sec, genre, release_date)`, `fact_viewing_session(session_id, profile_id, title_id, session_start_ts, watch_seconds, qualified_view_flag, country_code)`, `fact_production_spend(title_id, spend_date, amount_usd, category)`, `fact_payout(payout_id, talent_id, title_id, deal_id, payout_date, amount_usd)`.

SQL · 01 WINDOW · 15 MIN

For each series, find the episode in each season that had the highest 28-day completion rate. Return series_id, season_id, ep_num, completion_rate.

▸ SOLUTION

SQL · 02 GAPS & ISLANDS · 20 MIN

A "binge sequence" for a profile is a run of consecutive episodes within the same series watched within 24 hours of each other (in release order). Given `fact_viewing_session`, identify each binge sequence and return profile_id, series_id, sequence_start_ts, sequence_end_ts, episode_count.

▸ SOLUTION

SQL · 03 WINDOW · LAG · 15 MIN

For each title, calculate the "second-week drop-off" — the percentage decrease in qualified views from the first 7 days post-release to days 8-14. Return title_id, week1_views, week2_views, drop_pct.

▸ SOLUTION

SQL · 04 MULTI-CTE · FINANCE · 25 MIN

Calculate "viewing efficiency" per title in its first 90 days: total qualified views divided by production cost (USD). Return the top 10, but exclude titles with less than $1M production cost (too noisy) and less than 90 days since release.

▸ SOLUTION

SQL · 05 SCD2 QUERY · 20 MIN

Given **`dim_deal(deal_id, talent_id, title_id, base_fee_usd, effective_from, effective_to)`** with SCD2 (effective_to NULL = current), compute total contracted fee per talent as of any given date `D`. Write a query parameterized by `:as_of_date`.

▸ SOLUTION

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§ 05 — PYTHON DRILLS

Three problems. *25 minutes* each. Type out loud.

CodeSignal-style. The trick isn't the algorithm — it's getting through the boilerplate (parsing input, defaultdicts, edge cases) without hesitation. Set a timer. Talk while you type, like you would in the live round.

CODE · 01 SESSIONIZATION · 25 MIN

Given a list of viewing events `[(profile_id, title_id, ts, watch_seconds), ...]` (any order), group consecutive events for the same profile into "viewing sessions" where the gap between events is ≤ 30 minutes. Return a list of sessions: `[(profile_id, session_start, session_end, total_watch_seconds, distinct_titles)]`.

▸ SOLUTION

CODE · 02 SLIDING WINDOW · HEAP · 25 MIN

Implement a real-time "trending titles" tracker. Events arrive in chronological order: `(ts, title_id, watch_seconds)`. At any point you must be able to return the top K titles by watch_seconds *in the last hour*. Implement `add_event` and `top_k` with reasonable performance.

▸ SOLUTION

CODE · 03 HASH · AGGREGATION · 25 MIN

You have `views: List[(profile_id, title_id, watch_seconds)]` and `cohorts: Dict[profile_id, str]` mapping profiles to a cohort label (e.g., "new", "returning", "churned"). Compute, for each (cohort, title_id), total watch_seconds and unique viewer count. Return as a sorted list of `(cohort, title_id, watch, viewers)` sorted by watch descending.

▸ SOLUTION

The CodeSignal warm-up routine

The 5 minutes before the timer starts in CodeSignal: read the problem twice. Identify input/output types. Mentally run the example by hand. **Decide on data structures before typing.** "I need fast lookup — dict. I need ordered iteration — sort first. I need top-K — heap." 30 seconds of structure-picking saves 5 minutes of debugging.

§ 06 — STREAMING Q&A

Streaming concepts — answer each *out loud*.

90 seconds per answer. Talk to the wall. Then read the model answer. Repeat any you fumbled.

STREAM · 01

"What's the difference between event-time and processing-time? Why does it matter?"

▸ MODEL ANSWER (≈ 75 SEC)

STREAM · 02

"What is a watermark in Flink? How do you choose its value?"

▸ MODEL ANSWER

STREAM · 03

"Explain exactly-once semantics. How does Flink achieve it?"

▸ MODEL ANSWER

STREAM · 04

"What's the difference between Lambda and Kappa architectures? Which would you pick today?"

▸ MODEL ANSWER

STREAM · 05

"How do you handle schema evolution in a streaming system?"

▸ MODEL ANSWER

STREAM · 06

"Walk me through how a play event travels from a user's TV to the executive dashboard."

▸ MODEL ANSWER (THE 8-STEP NARRATIVE)

STREAM · 07

"What's backpressure? How do streaming systems handle it?"

▸ MODEL ANSWER

STREAM · 08

"Tumbling, hopping, and session windows — when do you use each?"

▸ MODEL ANSWER

§ 07 — ONE-PAGE CHEATSHEET

The page you read on day 5.

Print this. One side. Read it twice the morning of the interview, then put it away.

Modeling — the five moves

MOVE	WHAT YOU SAY / DO
1. Clarify	3-5 questions before drawing. Use cases, SLA, retention, key stability, time zone.
2. Grain	One sentence per fact: "One row per X per Y per Z." Out loud.
3. Conceptual → physical	Entities first. Then partitioning, clustering, SCD strategy.
4. SCDs & time	Type 1 (overwrite), Type 2 (versioned rows), Type 6 (hybrid). Justify.
5. Stress test	Volunteer 2 failure modes. Late data, GDPR, schema evolution, hot partitions.

SQL — the seven patterns

PATTERN	SHAPE

Top-N per group	ROW_NUMBER() OVER (PARTITION BY g ORDER BY metric DESC) → WHERE rn ≤ N
Gaps & islands	LAG → flag boundary → cumulative SUM = session_id → GROUP BY
Funnel	MAX(CASE WHEN step='X' THEN 1 END) per user → ratios with NULLIF
Dedup	ROW_NUMBER PARTITION BY natural_key ORDER BY ingest_ts DESC = 1
Cohort retention	cohort_week × activity_week LEFT JOIN, weeks_since = DATEDIFF
Running total	SUM() OVER (ORDER BY ... ROWS UNBOUNDED PRECEDING)
HLL / approx	APPROX_COUNT_DISTINCT – mention it for scale

Python — the five reflexes

NEED	REACH FOR
Group + aggregate	collections.defaultdict(int / list / set)
Top-K	heapq.nlargest, or manual min-heap of size K
Sliding window	collections.deque + running totals dict
Sessionization	sort within group, walk pairs, track open session, append on close
O(1) cache	collections.OrderedDict, move_to_end / popitem(last=False)

Streaming — vocabulary to drop

TERM	ONE-LINER
Watermark	"Promise: all events with ts ≤ W have been seen."
Event-time	"When it happened (client clock); not when we processed it."
Exactly-once	"Checkpoint + transactional sinks. Expensive. Most pipelines use at-least-once + idempotent writes."
Backpressure	"Downstream slower than upstream — system pauses upstream automatically."
Kappa	"One streaming pipeline. Backfills via log replay. Default today."
Schema registry	"Centralized schema validation; backward/forward compatibility modes."
Tumbling vs. hopping vs. session	"Non-overlapping vs. overlapping fixed-size vs. gap-defined dynamic."

Studio/Content vocabulary

TERM	MEANING
Slate	Planned set of releases for a period
Greenlight	Formal approval to begin production
Tentpole	High-budget flagship release
P&A	Prints & Advertising — marketing spend

`Residuals`	Ongoing payments to talent post-release
`MFN clause`	Most-favored-nation: my terms match the best terms granted to anyone else
`Amortization`	Spreading content cost over expected useful life
`License window`	Period during which a title is available in a territory

The 90-second answer template (for deep-dive STAR)

1. **Situation** (15s): "We had a [type of] pipeline running [scale]; the issue was [observable symptom]."
2. **Task** (15s): "I needed to [decision / outcome you owned]."
3. **Action** (45s): "I [diagnosed by X], decided to [Y because Z trade-off], implemented by [details]."
4. **Result** (15s): "[measurable outcome] within [timeframe]; [what changed structurally]."

What to do in the first 30 seconds of every problem

1. Restate the problem in your own words.
2. Ask 1-2 clarifying questions, even if obvious.
3. State assumptions out loud.
4. Walk through one example by hand.
5. Then start writing.

> THE SINGLE BIGGEST SIGNAL
>
> **Narrate. Constantly.** "I'm using a CTE here because the alternative is repeating this aggregation." "I'm picking ROW_NUMBER over RANK

> because ties shouldn't both qualify." Your interviewer cannot give you a senior score for code they only saw appear in silence.
